

SHUAI ZHOU

Master Student of Robotics

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EDUCATION

CARNEGIE MELLON UNIVERSITY

Master of Science in Robotics

Advisor: Zackory Erickson

Pittsburgh, USA

Aug 2026 — Jun 2028

SOUTH CHINA UNIVERSITY OF TECHNOLOGY

Bachelor of Engineering in Robotics; GPA: 3.86/4.00

Advisor: Jiaoyang Li

Guangzhou, China

Aug 2022 — Jun 2026

CARNEGIE MELLON UNIVERSITY

Visiting Student in the Robotics Institute

Advisor: Jiaoyang Li, Ding Zhao

Pittsburgh, USA

Jun 2025 — Mar 2026

UNIVERSITY OF CALIFORNIA, BERKELEY

Exchange Student in EECS; GPA: 4.00/4.00

Berkeley, USA

Aug 2023 — Dec 2023

EXPERIENCE

CARNEGIE MELLON UNIVERSITY, RCHI Lab

Research Topics: Dexterous Manipulation, Tool Use.

Advisor: Zackory Erickson

Pittsburgh, USA

Aug 2026 — Present

- Continual learning with failure-aware dexterous manipulation for tool use.

HKUST (Guangzhou), IPRN Lab

Research Topics: Foundation Model, Active Perception, Egocentric Human Data.

Advisor: Haoang Li

Guangzhou, China

Mar 2026 — Jul 2026

- Scaling Active Perception VLA with AMP teleoperation system and Human-Robot co-training [1].
- World action model pretraining with egocentric-human data and corresponding annotation pipeline [2] [5].

CARNEGIE MELLON UNIVERSITY, SAFE AI Lab

Research Topics: Humanoid, Egocentric Human Data, Whole-Body Controller.

Advisor: Ding Zhao

Pittsburgh, USA

Aug 2025 — Mar 2026

- Learning versatile humanoid manipulation policy and developing robust lower-body controller [7].
- Egocentric cross-embodiment manipulation with embodiment-specific latent prediction [4].

CARNEGIE MELLON UNIVERSITY, ARCS Lab

Research Topics: Learning for Planning, Multi-Robot System.

Advisor: Jiaoyang Li

Pittsburgh, USA

Apr 2025 — Mar 2026

- Collaborative multi-robot non-prehensile manipulation via flow-matching model and lazy planning [6].
- Learning execution-time prediction models for closed-loop multi-robot coordination and dependency optimization [3].

SHANGHAI JIAO TONG UNIVERSITY, RAP Lab

Research Topics: Heuristic Search, Multi-Robot System.

Advisor: Zhongqiang Ren

Shanghai, China

Apr 2024 — Apr 2025

- Scalable planners for multi-robot path planning that scale to 1,000+ robots [9].
- Complete, anytime planning framework with pruning and sorting strategies that accelerated search [8].

PUBLICATION

PREPRINT

- [1] **ActiveScale: Scaling Active Perception for Robots across Model, Data, and Hardware**
Shuai Zhou*, Kaisheng Pang*, Wenxuan Song, Wenjie Zhang, Xinhui Zheng, Haoang Li. **Under Review, 2026** [Paper]
- [2] **UniWAM: Unified World-Action Model**
Wenxuan Song, Jiayi Chen, Jingbo Wang, Shuai Zhou, Xicheng Gong, Zehua Fan, Ziyang Zhou, Junwu E, Wen Chen, Shunbo Zhou, Haoang Li. **Under Review, 2026**
- [3] **From Discrete Plans to Real-World Execution: A World-Model-Driven Framework for Execution-Aware Multi-Agent Path Finding**
Jingtian Yan, Shuai Zhou, He Jiang, Stephen Smith, Jiaoyang Li. **Under Review, 2026** [Paper]
- [4] **EgoX: Egocentric Cross-Embodiment Manipulation with Embodiment Dreaming**
Yaru Niu*, Binghong Chen*, Changwei Yao*, Shangtao Li*, Zhenlong Fang*, Revanth Krishna Senthilkumaran, Shuai Zhou, Yuemin Mao, Kalie Ching, Po-Ju Chao, Yudong Luo, Peiqi Yu, Heng Zhang, Hao Zhang, Bingqing Chen, Chen Qiu, Hongtei (Eric) Tseng, Changliu Liu, Jonathan Francis, Ding Zhao. **Under Review, 2026** [Paper]
- [5] **EgoANT: Segmentation and Annotation for Egocentric Human Video with open-source VLM**
Shuai Zhou **Blog, 2026** [Blog]

PUBLISHED

- [6] **LAMP: Long-Horizon Adaptive Manipulation Planning for Multi-Robot Collaboration in Cluttered Space**
Shuai Zhou, Yorai Shaoul, Jiaoyang Li. **IROS, 2026**. [\[Paper\]](#) [\[Project\]](#)
- [7] **HTD: Learning Versatile Humanoid Manipulation with Touch Dreaming**
Yaru Niu, Zhenlong Fang, Binghong Chen, Shuai Zhou, Revanth Krishna Senthilkumaran, Hao Zhang, Bingqing Chen, Chen Qiu, Hongtei (Eric) Tseng, Jonathan Francis, Ding Zhao. **IROS, 2026**. [\[Paper\]](#) [\[Project\]](#)
- [8] **LSRP*: Scalable and Anytime Planning for Multi-Agent Path Finding with Asynchronous Actions**
Shuai Zhou, Shizhe Zhao, Zhongqiang Ren. **SoCS, 2025**. [\[Paper\]](#) [\[Project\]](#)
- [9] **Loosely Synchronized Rule-Based Planning for Multi-Agent Path Finding with Asynchronous Actions**
Shuai Zhou, Shizhe Zhao, Zhongqiang Ren. **AAAI, 2025**. [\[Paper\]](#) [\[Project\]](#)

SKILLS

Platforms: Wuji Hand 2, AgileX ALOHA, AgileX Piper Arm, Apple Vision Pro, Meta Quest 2, Unitree G1 Humanoid

Simulators: MJLab, IsaacLab, MuJoCo, PyBullet

Programming: Python, C++, Java, HTML, MATLAB, L^AT_EX

SERVICES

Conference Reviewer: 2025-2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Journal Reviewer: 2026 IEEE Robotics and Automation Letters (RA-L)

HONORS

Merit Student of South China University of Technology

Top student in Robotics Engineering, Class of 2022

Guangzhou, China

Feb 2024